



MATHEMATICS

0580/43

Paper 4 (Extended)

May/June 2019

MARK SCHEME

Maximum Mark: 130

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **8** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Abbreviations

cao correct answer only

dep dependent

FT follow through after error

isw ignore subsequent working

oe or equivalent

SC Special Case

nfwf not from wrong working

soi seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	6h 27 mins	2	B1 for answerh 27 mins
1(a)(ii)	150 km/h	3	M2 for $\frac{90}{36} \times 60$ or M1 for $\frac{90}{\text{their time}}$ or B1 for 36 [mins] seen
1(a)(iii)	780	4	M3 for $\left(90 \times \frac{35}{3600}\right) \times 1000 - 95$ oe or M2 for $\left(90 \times \frac{35}{3600}\right) \times 1000$ oe or B1 for figs 875 or M1 for $90 \times \frac{35}{3600}$ seen or for $90 \times \frac{1000}{3600}$ oe If 0 scored, SC1 for <i>their</i> distance (> 95) – 95
1(b)(i)	7 : 5	1	
1(b)(ii)	66.7 or 66.66 to 66.67	3	M2 for $\frac{140 - 84}{84} [\times 100]$ oe or for $\frac{140}{84} \times 100$ oe or M1 for $\frac{140}{84}$ oe
1(b)(iii)	24 576	5	M4 for complete method, $40 \times 60 + 0.7 \times 220 \times 84 + 0.3 \times 220 \times 140$ oe OR B1 for 40 [children] M1 for $0.7 \times 220 \times 84$ oe M1 for $0.3 \times 220 \times 140$ oe B1 for 2400 or 12936 or 9240 nfwf

Question	Answer	Marks	Partial Marks
1(c)	3.5×10^5 nfw	3	M2 for $3.08 \times 10^5 \div \left(\frac{100-12}{100}\right)$ oe or M1 for $3.08 [\times 10^5]$ associated with (100–12)%
2(a)	–10	2	M1 for $-17 - 3 = 7x - 5x$ oe or better
2(b)	–1, 0, 1, 2 final answer	3	B2 for 3 correct values and no incorrect values or 4 correct values and one incorrect value or M2 for $-\frac{7}{4} < n \leq 2$ oe or M1 for $-\frac{7}{4} < n \leq k$ or $k < n \leq 2$ oe
2(c)(i)	a^9	1	
2(c)(ii)	$125x^3y^6$ final answer	2	B1 for 2 correct elements if in form kx^ny^m
2(c)(iii)	$\frac{4y^{[1]}}{3x^4}$ final answer	3	B2 for $\left(\frac{3x^4}{4y^{[1]}}\right)^{[-1]}$ oe seen OR B1 for $3x^4$ or $4y^{[1]}$ and M1 for $\left(\frac{64y^3}{27x^{12}}\right)^{[\frac{1}{3}]}$ oe If 0 scored, SC1 for $\frac{64y^{[1]}}{27x^4}$ or $\frac{0.333x^{-4}}{0.25y^{-1}}$ seen
3(a)(i)	Image at (–5, 4), (–2, 4), (–4, 6)	2	B1 for translation by $\begin{pmatrix} -3 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ 2 \end{pmatrix}$
3(a)(ii)	Image at (2, 1), (4, –1), (2, –2)	2	B1 for reflection in $y = -x$ or $y = x$ drawn
3(b)	Rotation 90°[anticlockwise] oe (1, –1)	3	B1 for each
3(c)(i)	$\begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$	2	B1 for 2 by 2 matrix with one correct row or column
3(c)(ii)	Strict FT <i>their</i> (c)(i)	1	Answer not equal to zero FT <i>their</i> (c)(i) only if 2 by 2

Question	Answer	Marks	Partial Marks
4(a)(i)	$\frac{1}{2} \times \frac{4}{3} \times \pi \times 5.6^3$	M1	
	367.8... to 367.9	A1	
4(a)(ii)	3.06 or 3.060 to 3.061...	4	M1 for $0.8 \times 368 [= 294.4]$ M2 for $[r^2 =] \frac{\text{their } 294.4}{10\pi}$ oe or M1 for $\pi r^2 \times 10 = \text{their } 294.4$ oe
4(b)(i)	44[.0] or 43.98 to 43.99 nfw	5	B2 for [slant height =] $\frac{25}{4}$ oe or M1 for $[l^2 =] 6^2 + 1.75^2$ oe M2 for $\pi \times 1.75 \times \text{their } l + \pi \times 1.75^2$ or M1 for $\pi \times 1.75 \times \text{their } l$ or $\pi \times 1.75^2$
4(b)(ii)(a)	SF = $\frac{1}{4}$ oe soi	B1	
	$\frac{1}{3} \pi \times 1.75^2 \times 6 - \frac{1}{3} \pi \times \text{their } 0.4375^2 \times 1.5$ OR $\frac{1}{3} \pi \times 1.75^2 \times 6 \times \left(1 - \left(\frac{1}{4}\right)^3\right)$ oe	M2	M1 for $\frac{1}{3} \pi \times 1.75^2 \times 6$ or $\frac{1}{3} \pi \times \text{their } 0.4375^2 \times 1.5$ OR M1 for $1 - \left(\frac{1}{4}\right)^3$ oe
	18.94 or 18.939 to 18.944...	A1	
4(b)(ii)(b)	95 final answer	3	B2 for 94.5 or 94.69 to 94.722 OR M2 for $18.9 \times 10^3 \div 200$ oe or M1 for 18.9×10^3 or $200 \div 10^3$ or figs 189.. $\div$ 200 or 18.9.. $\div$ figs 2
5(a)(i)	−3	1	
5(a)(ii)	6.2 to 6.4 oe	2	M1 for 3 seen or used
5(b)	$y = 5 - 3x$ ruled	2	B1 for $y = 5 - 3x$ soi or ruled line with gradient − 3 or with y – intercept at 5 (but not $y = 5$) or B1FT for incorrect line equation/expression shown in working and <i>their</i> line correctly drawn
	− 0.3 to − 0.2 1.65 to 1.8	2	B1 for each, dep on $y = 5 - 3x$ drawn or FT <i>their</i> line provided equation/expression shown in working, dep on B1FT for line

Question	Answer	Marks	Partial Marks
5(c)	Tangent ruled at $x = -2$	1	B1 for correct tangent
	-4.5 to -2.5	2	Dep on B1 for tangent or close attempt at tangent at $x = -2$ M1 for rise/run also dep on tangent drawn or close attempt at correct tangent Must see correct or implied calculation from a drawn tangent
5(d)(i)	8, 4, 0.25 oe	3	B1 for each
5(d)(ii)	Correct graph	3	B2FT for 6 or 7 correct plots or B1FT for 4 or 5 correct plots
5(d)(iii)	1.8 to 1.9	1	
6(a)	40.5 or 40.45[8..] or 40.46 nfw	4	M1 for 25, 32.5, 37.5, 50, 80 soi M1 for Σft M1 dep for their $\Sigma ft \div 120$
6(b)	Fully correct histogram	4	B1 for each correct bar If 0 scored, SC1 for frequency densities of 5.4, 4.2, 0.8 and 0.45 seen
7(a)	$[y =] 4x + 5$	3	B2 for answer $[y =] 4x + c$ oe (c can be numeric or algebraic) OR M2 for $\frac{y-9}{x-1} = \frac{9-(-3)}{1-(-2)}$ oe OR M1 for $\frac{9-(-3)}{1-(-2)}$ oe or for M1 for correct substitution of $(-2, -3)$ or $(1, 9)$ into $y = (their\ m)x + c$ oe
7(b)	76[.0] or 75.96...	2	M1 for $\tan[] = 4$ oe

Question	Answer	Marks	Partial Marks
7(c)(i)	$[y =] -\frac{1}{4}x + \frac{23}{8}$ oe	3	B2FT for $[y =] -\frac{1}{\text{their gradient from (a)}}x + c$ oe (c can be numeric or algebraic) OR M2 for $\frac{y-2}{x-3.5} = -\frac{1}{\text{their gradient from (a)}}$ oe OR M1 for $-\frac{1}{\text{their gradient from (a)}}$ soi M1 for correct substitution of (3.5, 2) into $y = (\text{their } m)x + c$ oe
7(c)(ii)	(-4.5, 4)	2	B1 for each value or for $\begin{pmatrix} -8 \\ 2 \end{pmatrix}$ seen
8(a)(i)	$\frac{x-1}{x+2}$	2	B1 for either numerator or denominator correct
8(a)(ii)(a)	$\frac{x}{x+3} \times \frac{x-1}{x+2} = \frac{7}{15}$	B1	FT <i>their (a)(i)</i> = $\frac{7}{15}$
	$15x(x-1) = 7(x+3)(x+2)$	M1	Removes all algebraic fractions FT <i>their</i> equation if in comparable form
	$15x^2 - 15x = 7x^2 + 21x + 14x + 42$	M1	Correctly expands all brackets FT <i>their</i> equation if in comparable form
	$[8x^2 - 50x - 42 = 0]$ $4x^2 - 25x - 21 = 0$	A1	With no errors or omissions seen and one further stage seen after final M1
8(a)(ii)(b)	$(4x+3)(x-7) [= 0]$	M2	M1 for $4x(x-7) + 3(x-7)$ or $x(4x+3) - 7(4x+3)$ or for $(4x+a)(x+b)$ where either $ab = -21$ or $4b+a = -25$ If 0 scored, SC1 for $4x+3$ and $x-7$ seen but not in factorised form
	7 and $-\frac{3}{4}$	B1	
8(a)(ii)(c)	7	1	FT <i>their</i> positive solution

Question	Answer	Marks	Partial Marks
8(b)	$\frac{1}{6}$ oe	4	M3 for $\frac{5}{9} \times \frac{4}{8} \times \frac{3}{7} + \frac{4}{9} \times \frac{3}{8} \times \frac{2}{7}$ or M2 for $\frac{5}{9} \times \frac{4}{8} \times \frac{3}{7}$ or $\frac{4}{9} \times \frac{3}{8} \times \frac{2}{7}$ or M1 for $\frac{5}{9}, \frac{4}{8}, \frac{3}{7}$ seen or $\frac{4}{9}, \frac{3}{8}, \frac{2}{7}$ seen If 0 scored, SC1 for $\frac{5^3 + 4^3}{729}$ oe
9(a)(i)	$\angle ACD = 46$ soi or $\angle CDE = 44$ soi	B2	B1 for angle $ADC = 108$ or angle $DCB = 18$
	$\frac{58 \sin 108}{\sin \text{their} 46}$	M2	M1 for $\frac{\sin 108}{x} = \frac{\sin \text{their} 46}{58}$ oe
	76.68... nfww	A1	
9(a)(ii)	10.9 or 10.91 to 10.94	3	B2 for $[AB =] 68.9$ or 68.91 to 68.94 or M2 for a correct explicit statement for AB or BD or M1 for $\frac{AB}{76.7} = \cos 26$ oe
9(b)(i)	10.4 or 10.43 to 10.44	4	M3 for $\sqrt{\frac{70}{\sin 40}}$ oe or M2 for $x^2 \times \sin 40 = 70$ oe or M1 for $\frac{1}{2}x \times 2x \times \sin 40 = 70$
9(b)(ii)	140	1	
10(a)(i)	3, -1	2	B1 for each
10(a)(ii)	$23 - 4n$ oe final answer	2	M1 for $k - 4n$ or $23 - jn$ ($j \neq 0$)
10(a)(iii)	22	2	M1 for <i>their</i> (a)(ii) = -65
10(b)	23	2	B1 for 37 or 60